Task 1: Data Cleaning and Preprocessing Report

Data Masters Challenge 2026

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1 Overview

The original dataset contained **390 observations** and **17 variables**. The objective was to audit data quality, address missingness, correct impossible values, remove duplicates, assess outliers, engineer features, and produce a modeling-ready dataset.

After cleaning, the final dataset contained **387 observations** and **54 variables**.

Stage	Rows	Columns
Raw Data	390	17
After Duplicate Removal	387	20
Final Clean Dataset	387	54

Table 1: Before vs After Summary

2 Missing Value Analysis

Missingness included both ordinary missing data and systematic sport-specific missingness.

Variable	Missing %	Type
Medal	73.3	Structural (No Medal)
Reaction Time	26.2	MNAR
World Cup Points	25.1	MNAR
Body Fat	8.0	MCAR/MAR
VO2max	7.2	MCAR/MAR
Training Hours	6.9	MCAR/MAR

Table 2: Key Missingness Patterns

MNAR vs MCAR. Reaction Time and World Cup Points were missing for systematic sport-specific reasons (e.g., Curling, Figure Skating, Ice Hockey), indicating MNAR missingness rather than random loss. Missingness indicators were retained as predictive features.

Numeric missing values were imputed using:

- Sport + Gender median
- Sport median fallback
- Global median fallback

Medal missingness was recoded as `None` rather than imputed.

3 Duplicate and Error Correction

Duplicates were detected using all substantive athlete attributes excluding Athlete ID. Three duplicate records were identified and removed.

Four impossible values were corrected:

Rather than deleting observations, impossible entries were set to missing and imputed using sport-gender medians. Error flags were retained.

Variable	Error	Action
Age	9	Imputed
Training Hours	172	Imputed
Body Fat	99.2	Imputed
VO2max	-8.4	Imputed

Table 3: Impossible Values Corrected

4 Outlier Analysis

Outliers were evaluated using IQR and treated cautiously to distinguish errors from genuine elite extremes.

Variable	Outliers	%
Body Fat	14	3.6
Reaction Time	15	3.9
World Cup Points	7	1.8
Tradition Index	35	9.0

Table 4: IQR Outlier Summary

No legitimate extremes were removed. Notably, the 42-year-old Olympic debut figure skater was retained as a genuine observation, illustrating why outliers should not be blindly discarded.

5 Feature Engineering

Three predictive features were engineered:

Feature	Purpose
VO2/Body Fat Efficiency	Aerobic efficiency measure
Experience Training Index	Experience × workload interaction
GDP Tradition Index	National support + sport culture signal

Table 5: Engineered Features

These features incorporate physiological, training, and country-level interactions not captured by raw variables alone.

6 Final Clean Dataset

Final cleaning decisions included:

- Removed 3 duplicate records
- Corrected 4 impossible values
- Imputed missing numeric values using hierarchical medians
- Preserved MNAR structure via indicator variables
- Retained genuine outliers
- Added engineered predictive features

The final modeling-ready dataset contains **387 rows** and **54 variables**.

7 Conclusion

The cleaning process balanced statistical rigor with domain knowledge. Data errors were corrected, genuine extremes preserved, sport-specific missingness explicitly modeled, and informative features engineered.

The resulting dataset is cleaner, interpretable, and suitable for downstream predictive modeling. Below is the table showing the before vs after summary:

Stage	**Rows**	**Columns**	**Notes**
Raw dataset	390	17	Original dataset
After duplicate removal	387	20	Removed duplicate athlete records based on all non-ID fields
After impossible values set to missing	387	20	Impossible values flagged and replaced with missing values prior to imputation
After imputation and feature engineering	387	54	Missing values imputed; engineered features and indicator flags added

Table 6: Before vs. After Cleaning Summary